

Investments I Final Review Slides

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Final Exam Format & Logistics

- Mark your calendars: **Thursday, May 7th, 9:00 AM- 12:00 PM in Alumnae Hall (North Campus)**
- You have 3 hours to take an exam designed for 2 hours...less time pressure!
 - Two Parts:
 - New Material: Chapters 18, 20, 21 (Equity Valuation and Options)
 - Cumulative: Everything you learned before Midterm 2!
 - Chapters 1-7, 9, 11, 14
- Review your past exams for practice!
- Go through the old midterm review slides (posted on Canvas)
- The best way to study is doing the 10 homeworks

Chapter 18: Equity Pricing



Chapter 18 Key Ideas

- A stock's intrinsic value is the present value of all future payments
- Market capitalization rate (k) = return required by investors
- Multi-stage DDM
- Dividend growth rate (g), plowback ratio (b)
- In equilibrium, $E(r) = k$
- The key to doing well on these questions is understanding how to apply the correct formulas

Chapter 18 Key Formulas

$$k = r_f + \beta(E(r_m) - r_f)$$

$$E(r) = \frac{D_1 + P_1 - P_0}{P_0} = \frac{D_1}{P_0} + g$$

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k)^t} = \sum_{t=1}^H \frac{D_t}{(1+k)^t} + \frac{P_H}{(1+k)^H}$$

$$k = \frac{D_1}{P_0} + g$$

$$V_0 = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+k)^t} = \frac{D_0(1+g)}{k-g} = \frac{D_1}{k-g}$$

$$P_1 = P_0(1+g)$$



Complications to Constant Growth DDM

Reinvesting earnings

- Dividend payout ratio + plowback ratio = 1

$$\frac{D_1}{E_1} + b = 1$$

- Growth rate changes as plowback ratio changes $g = \text{ROE} * b$
- Intrinsic value is no growth value per share + present value of growth opportunities

$$V_0 = \frac{E_1}{k} + PVGO$$



Complications to Constant Growth DDM

Multistage DDM

- **Two Stage DDM:** Where T is the time we hit the steady growth phase,

$$V_0 = \sum_{t=0}^T \frac{D_t}{(1+k)^t} + \frac{D_T(1+g)}{k-g}$$

- Can extend to



Extensions of DDM

Price-Earnings Ratio

- Price per share / earnings per share can act as an estimate for g

Free Cash Flow Evaluation

- Value firm based on discounting free cash flow rather than dividends

$$\text{Firm value} = \sum_t \frac{FCFF_t}{(1+WACC)^t} + \frac{V_T}{(1+WACC)^T}$$

$$\text{Intrinsic value to equity} = \sum_t \frac{FCFE_t}{(1+k_E)^t} + \frac{V_T}{(1+k_E)^T}$$

FCFF - free cash flow to firm; WACC - weighted average cost of capital ; FCFE - free cash flow to equity



Chapter 18 Practice Problems

1. Music Doctors Company has an expected ROE of 14%. The dividend growth rate will be _____ if the firm follows a policy of paying 60% of earnings in the form of dividends.
 - A. 4.8%
 - B. 5.6%
 - C. 7.2%
 - D. 6.0%
 - E. none of the above

2. If a firm has a required rate of return equal to the ROE
 - A. the firm can increase market price and P/E by retaining more earnings.
 - B. the firm can increase market price and P/E by increasing the growth rate.
 - C. the amount of earnings retained by the firm does not affect market price or the P/E.
 - D. A and B.
 - E. none of the above.



Chapter 18 Practice Problems

1. Music Doctors Company has an expected ROE of 14%. The dividend growth rate will be _____ if the firm follows a policy of paying 60% of earnings in the form of dividends.

A. 4.8%

B. 5.6% → $14\% \times 0.40 = 5.6\%$.

C. 7.2%

D. 6.0%

E. none of the above

2. If a firm has a required rate of return equal to the ROE

A. the firm can increase market price and P/E by retaining more earnings.

B. the firm can increase market price and P/E by increasing the growth rate.

C. the amount of earnings retained by the firm does not affect market price or the P/E.

D. A and B.

E. none of the above.



Chapter 18 Practice Problems

3. A company paid a dividend last year of \$1.75. The expected ROE for next year is 14.5%. An appropriate required return on the stock is 10%. If the firm has a plowback ratio of 75%, the dividend in the coming year should be

- A. \$1.80
- B. \$2.12
- C. \$1.77
- D. \$1.94
- E. none of the above

4. You wish to earn a return of 13% on each of two stocks, X and Y. Stock X is expected to pay a dividend of \$3 in the upcoming year while Stock Y is expected to pay a dividend of \$4 in the upcoming year. The expected growth rate of dividends for both stocks is 7%. The intrinsic value of stock X _____.

- A. will be greater than the intrinsic value of stock Y
- B. will be the same as the intrinsic value of stock Y
- C. will be less than the intrinsic value of stock Y
- D. A or B
- E. none of the above is a correct answer.



Chapter 18 Practice Problems

3. A company paid a dividend last year of \$1.75. The expected ROE for next year is 14.5%. An appropriate required return on the stock is 10%. If the firm has a plowback ratio of 75%, the dividend in the coming year should be

A. \$1.80

B. \$2.12

C. \$1.77

D. \$1.94 -> $g = .145 \times .75 = 10.875\%$; $\$1.75(1.10875) = \1.94 .

E. none of the above

4. You wish to earn a return of 13% on each of two stocks, X and Y. Stock X is expected to pay a dividend of \$3 in the upcoming year while Stock Y is expected to pay a dividend of \$4 in the upcoming year. The expected growth rate of dividends for both stocks is 7%. The intrinsic value of stock X _____.

A. will be greater than the intrinsic value of stock Y

B. will be the same as the intrinsic value of stock Y

C. will be less than the intrinsic value of stock Y

D. A or B

E. none of the above is a correct answer.

→ $PV_0 = D_1 / (k - g)$; given k and g are equal, the stock with the larger dividend will have the higher value.



Chapter 18 Practice Problems (SA)

5. Low Tech Chip Company is expected to have EPS of \$2.50 in the coming year. The expected ROE is 14%. An appropriate required return on the stock is 11%. Suppose the firm has a dividend payout ratio of 40%.

(a) What is the dividend growth rate (2 points)?

(b) What is the intrinsic value of the stock (4 points)?



Chapter 18 Practice Problems (SA)

5. Low Tech Chip Company is expected to have EPS of \$2.50 in the coming year. The expected ROE is 14%. An appropriate required return on the stock is 11%. Suppose the firm has a dividend payout ratio of 40%.

(a) What is the dividend growth rate (2 points)?

$$g = 14\% \times 0.6 = 8.4\%$$

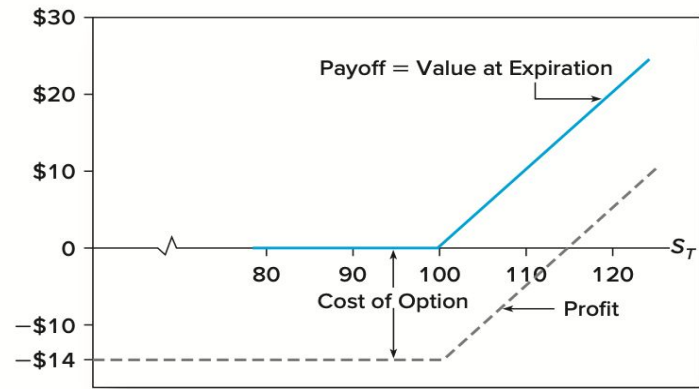
(b) What is the intrinsic value of the stock (4 points)?

$$D_1 = \$2.50(0.4) = \$1.00 \dots P = 1 / (.11 - 0.084) = \$38.46$$

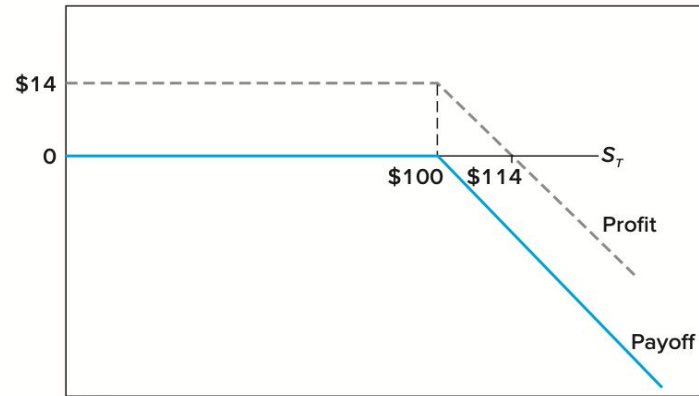
Chapter 20+21: Options

Payoffs - Calls

- Payoff to call holder = $\max(S_T - X, 0)$
 - Profit to call holder = $\max(S_T - X, 0) - P$
- Payoff to call writer = $\min(-(S_T - X), 0)$
 - Profit to call writer = $\min(-(S_T - X), 0) + P$



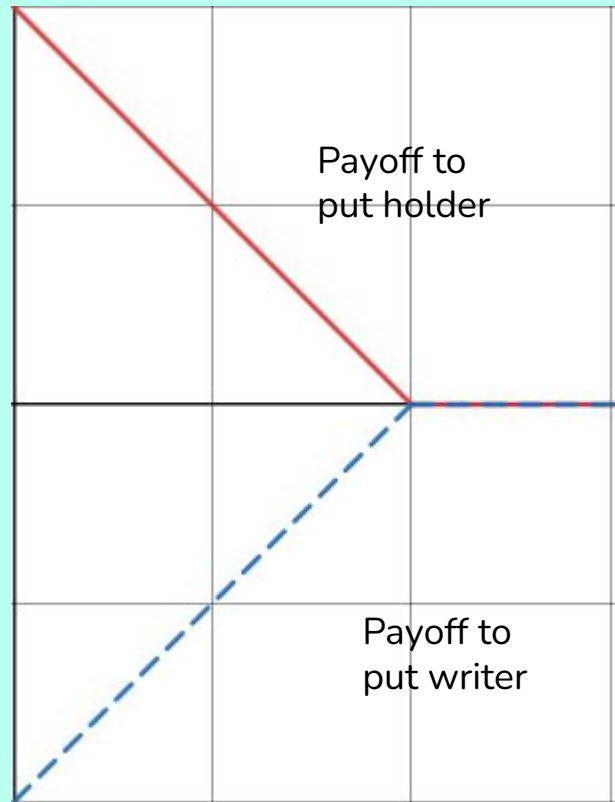
Payoff and Profit to Call Holder



Payoff and Profit to Call Writer

Payoffs - Puts

- Payoff to put holder = $\max(X - S_T, 0)$
 - Profit to put holder = $\max(X - S_T, 0) - P$
- Payoff to put writer = $\min(-(X - S_T), 0)$
 - Profit to put writer = $\min(-(X - S_T), 0) + P$





Put-Call Parity

- For European options with no dividends (Add $\text{Div}/(1+r_f)^T$ to the left-hand side for American options)

$$C + \frac{X}{(1+r_f)^T} = S_0 + P$$

- If put-call parity violated, arbitrage opportunities:
 - Buy the cheap portfolio, sell the expensive portfolio



Types of Options

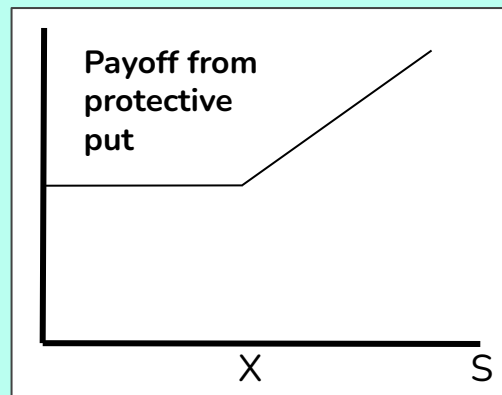
- **Naked option** - write without holding underlying asset, most puts
- **Covered/married option** - write while holding underlying asset, most calls
- **In the money** - option should be exercised
- **Out of the money** - option should not be exercised
- **American option** - exercise at or before expiration
- **European** - exercise at expiration



Protective Put

- Simultaneously **buy a stock** and **buy a put** on the same stock
- Limit potential losses from stock drop in price

	Stock	Put	Combined
Payoff	S_T	$\max(X - S_T, 0)$	$\max(X, S_T)$
Cost	S_0	P	$S_0 + P$
Profit	$S_T - S_0$	$\max(X - S_T, 0) - P$	$\max(X, S_T) - S_0 - P$

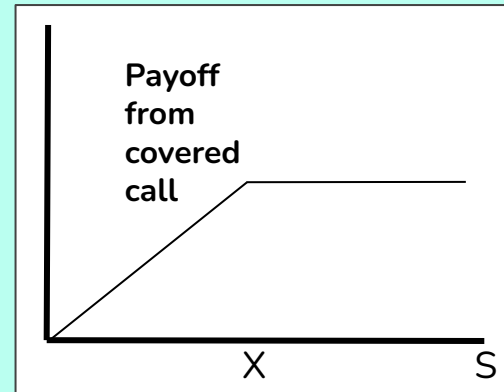




Covered Call

- Simultaneously **buy a stock** and **write a call** on the same stock
- Limit potential losses from exercise of call

	Stock	Call	Combined
Payoff	S_T	$\min(X - S_T, 0)$	$\min(X, S_T)$
Cost	S_0	$-C$	$S_0 - C$
Profit	$S_T - S_0$	$\min(X - S_T, 0) + C$	$\min(X, S_T) - S_0 + C$

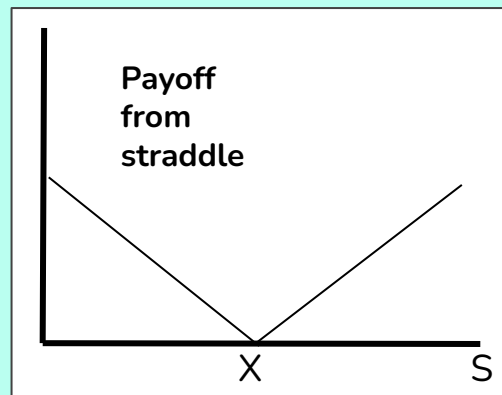




Straddle

- Simultaneously **buy a call** and **buy a put** on the same stock with the same X and T
- Believe stock price will move a lot but unsure about direction
- Strip - buy 2 puts + 1 call
- Strap - buy 2 calls + 1 put

	Call	Put	Combined
Payoff	$\max(S_T - X, 0)$	$\max(X - S_T, 0)$	$ S_T - X $
Cost	C	P	$C + P$
Profit	$\max(S_T - X, 0) - C$	$\max(X - S_T, 0) - P$	$ S_T - X - C - P$





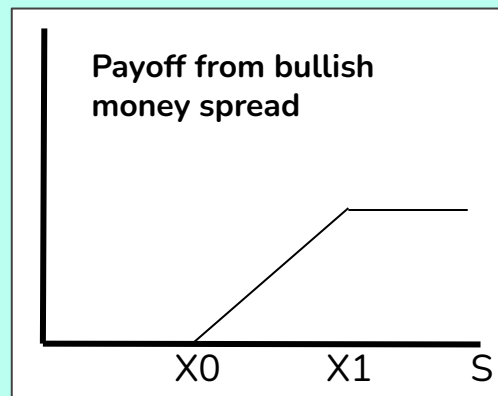
Spread

- Simultaneously **buy and sell call/put** with different X s (money spread) or T s (time spread)
- Bullish spread - think price will increase
- Bearish spread - think price will decrease

Bullish Money Spread

Buy call with X_0 , sell call with X_1 , $X_0 < X_1$

	$S_T < X_0$	$X_0 < S_T < X_1$	$X_1 < S_T$
Payoff	0	$S_T - X_0$	$X_1 - X_0$
Cost	$C_0 - C_1$	$C_0 - C_1$	$C_0 - C_1$
Profit	$-C_0 + C_1$	$S_T - X_0 - C_0 + C_1$	$X_1 - X_0 - C_0 + C_1$





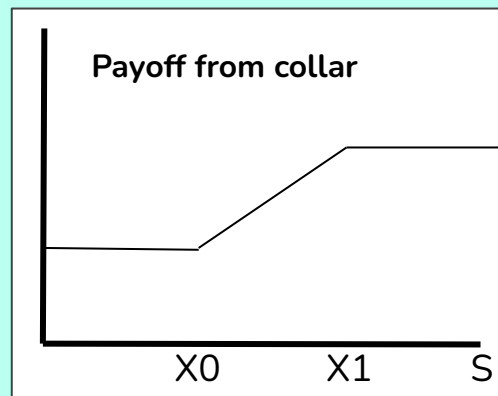
Collar

- Simultaneously **buy a put**, **buy the stock**, and **sell a call** on the same stock
- Brackets value of portfolio between two bounds

Collar

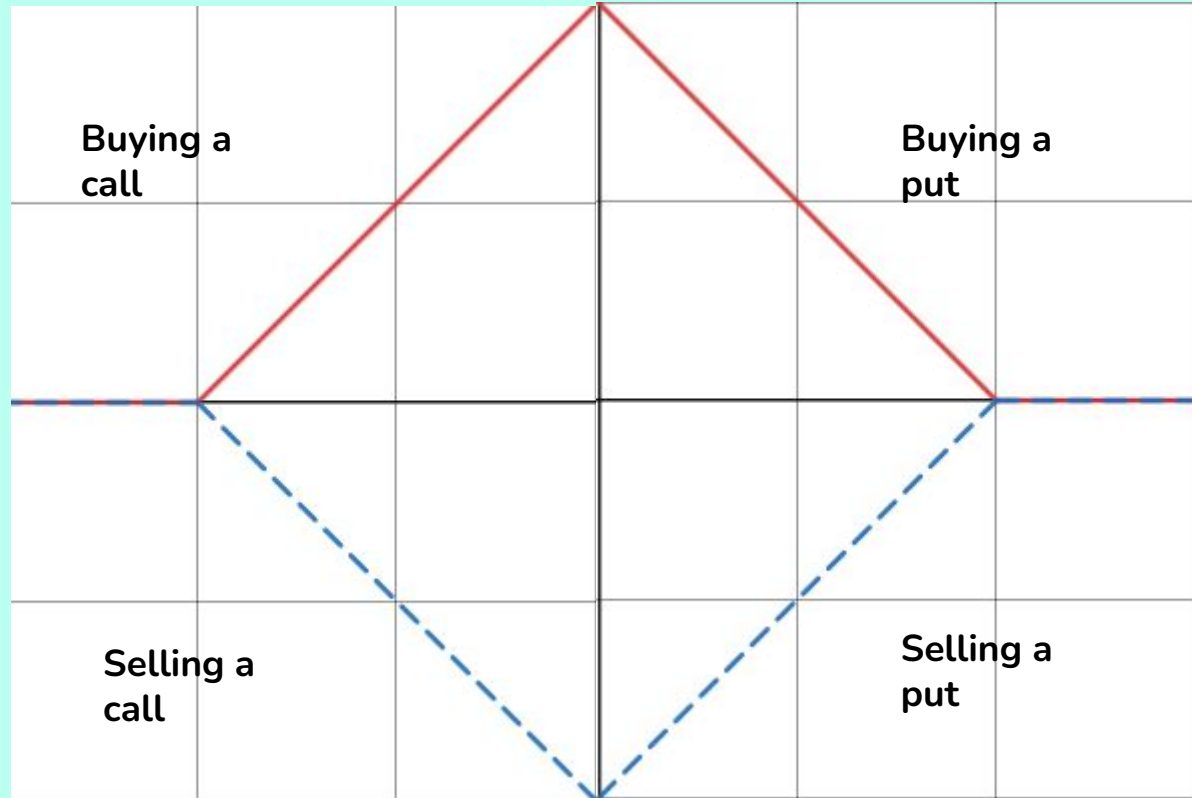
Buy put with X_0 , buy stock at S_0 , sell call with X_1 , $X_0 < X_1$

	$S_T < X_0$	$X_0 < S_T < X_1$	$X_1 < S_T$
Payoff	X_0	S_T	X_1
Cost	$P + S_0 - C$	$P + S_0 - C$	$P + S_0 - C$
Profit	$X_0 - P - S_0 + C$	$S_T - P - S_0 + C$	$X_1 - P - S_0 + C$





Tips for Drawing Option Strategy Payoffs





Tips for Drawing Option Strategy Payoffs

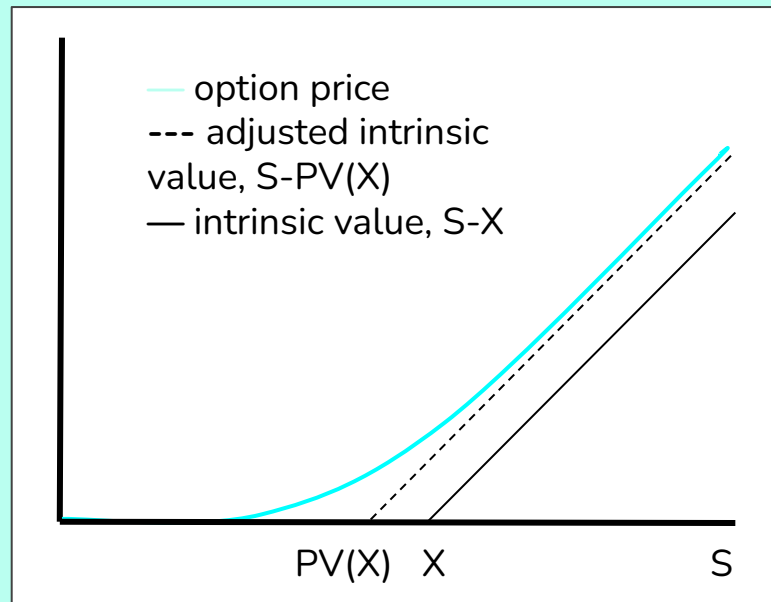
- Buying a stock always has +1 slope
- Selling a stock always has -1 slope
- Buying a call or selling a put have +1 slope for the range where the option is exercised, 0 otherwise
- Selling a call or buying a put have -1 slope for the range where the option is exercised, 0 otherwise



Option Value

- **Intrinsic value of an option:** value if option expired right now
 - Always positive or 0
- **Time value of an option:** option price - intrinsic value
 - Always positive or 0

Call Option Value Before Expiration





Impact on Call Option Value

- Stock price - positive correlation
- Exercise price - negative correlation
- Volatility - positive correlation
- Time to expiration - positive correlation
- Interest rate - positive correlation
- Dividend payout ratio - negative correlation



Impact on Put Option Value

- Stock price - negative correlation
- Exercise price - positive correlation
- Volatility - positive correlation
- Time to expiration - positive correlation
- Interest rate - negative correlation
- Dividend payout ratio - positive correlation



Restrictions on Option Values

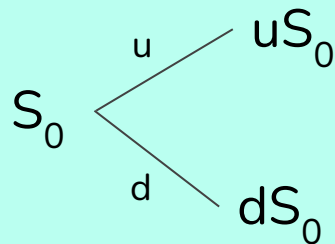
- For call option, right to exercise early is worthless, would rather sell option
- For non-dividend paying stock, value of American call = value of European call
- There is a value to early exercise for a put, so the value of an American put $>$ value of European put



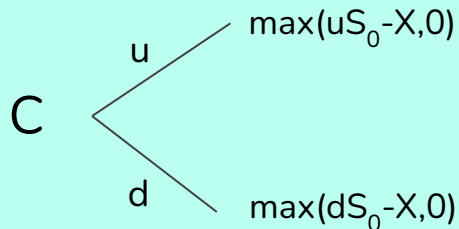
Two State Option Pricing

- Two state option pricing is an exceptionally simple case that lets us get the gist of pricing options without super complex math.
- Value tree:
 - Two time periods: $t=0$, $t=1$
 - At $t=1$, stock price can take two values
 - Downstate: dS_0
 - Update: uS_0
- Value tree to an option
 - We can calculate the intrinsic value (0 or spread between stock price and exercise price) of the option given the upstate and downstate stock prices.

Value Tree to a Stock



Value Tree to a Call





Perfect Hedge Strategy (Binomial Option Pricing)

- Given stock price and exercise price, construct payoffs
- Hedge ratio for a call $H = \frac{C_u - C_d}{uS_0 - dS_0}$ between 0 and 1
 - Buy H shares, write 1 call; certainty payoff the same in up and down scenarios
- Hedge ratio for a put $H = \frac{P_u - P_d}{uS_0 - dS_0}$ between -1 and 0
 - Buy H shares, buy 1 put; certainty payoff the same in up and down scenarios
- Value of hedged portfolio is PV(certainty payoff), isolate price of option



Example: Perfect Hedge

- $S_0 = 100$, $dS_0 = 80$, $uS_0 = 120$, call with $X = 100$, $r_f = 5\%$
- Calculate H
- Decide which assets to buy/sell
- Solve for certainty payoff
- Solve for PV(certainty payoff)
- Isolate value of call by equating cost of risk-free portfolio and hedged portfolio
- $H = (20-0) / (120-80) = \frac{1}{2}$
- **Buy $\frac{1}{2}$ stock, sell 1 call**
- $60-20 = \mathbf{40}$
- $40/(1.05) = \mathbf{38.10}$
- $50-C=38.10 \rightarrow C = \mathbf{11.90}$

If finding the value of a put, you will ADD P because you're buying a put, which increases cost.



Chapter 20+21 Practice Problems

6. Buyers of put options anticipate the value of the underlying asset will _____ and sellers of call options anticipate the value of the underlying asset will _____.

- A. increase; increase
- B. decrease; decrease
- C. decrease; increase
- D. increase; decrease
- E. cannot tell without further information

7. The potential loss for a writer of a naked call option on a stock is

- A. limited.
- B. unlimited.
- C. larger the lower the stock price.
- D. equal to the call premium.
- E. none of the above.



Chapter 20+21 Practice Problems

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- C. decrease; increase
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- D. equal to the call premium.
- E. none of the above.



Chapter 20+21 Practice Problems

8. Suppose you purchase one IBM May 100 call contract at \$5 (i.e. exercise price is \$100) and write one IBM May 105 call contract at \$2. The maximum potential profit of your strategy is _____ if both options are exercised.

- A. \$600.
- B. \$500.
- C. \$200.
- D. \$300.
- E. \$100

9. A protective put strategy is

- A. a long put plus a long position in the underlying asset.
- B. a long put plus a long call on the same underlying asset.
- C. a long call plus a short put on the same underlying asset.
- D. a long put plus a short call on the same underlying asset.
- E. none of the above.



Chapter 20+21 Practice Problems

8. Suppose you purchase one IBM May 100 call contract at \$5 (i.e. exercise price is \$100) and write one IBM May 105 call contract at \$2. The maximum potential profit of your strategy is _____ if both options are exercised.

A. \$600.

B. \$500.

Profit Breakdown:

C. \$200. → **Profit from exercising held Call:** $S_T - \$100 - \5

D. \$300. **Profit when written Call is exercised:** $-(S_T - \$105) + \2

E. \$100 **Net Profit** = $[(S_T - \$100) - \$5] + [-(S_T - \$105) + \$2] = \$2 \rightarrow \$2 * 100 \text{ shares} = \200

9. A protective put strategy is

A. a long put plus a long position in the underlying asset.

B. a long put plus a long call on the same underlying asset.

C. a long call plus a short put on the same underlying asset.

D. a long put plus a short call on the same underlying asset.

E. none of the above.

→ **If you invest in a stock and purchase a put option on the stock, you are guaranteed a payoff equal to the exercise price; thus the protection of the put.**



Chapter 20+21 Practice Problems

10. If the stock price increases, the price of a put option on that stock _____ and that of a call option _____.
- A. increases, decreases
 - B. increases, increases
 - C. decreases, increases
 - D. decreases, decreases
 - E. does not change, does not change
11. Other things equal, the price of a stock call option is positively correlated with the following factors except
- A. the stock price.
 - B. the time to expiration.
 - C. the stock volatility.
 - D. the exercise price.
 - E. none of the above.



Chapter 20+21 Practice Problems

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- A. increases, decreases
- B. increases, increases
- C. decreases, increases**
- D. decreases, decreases
- E. does not change, does not change

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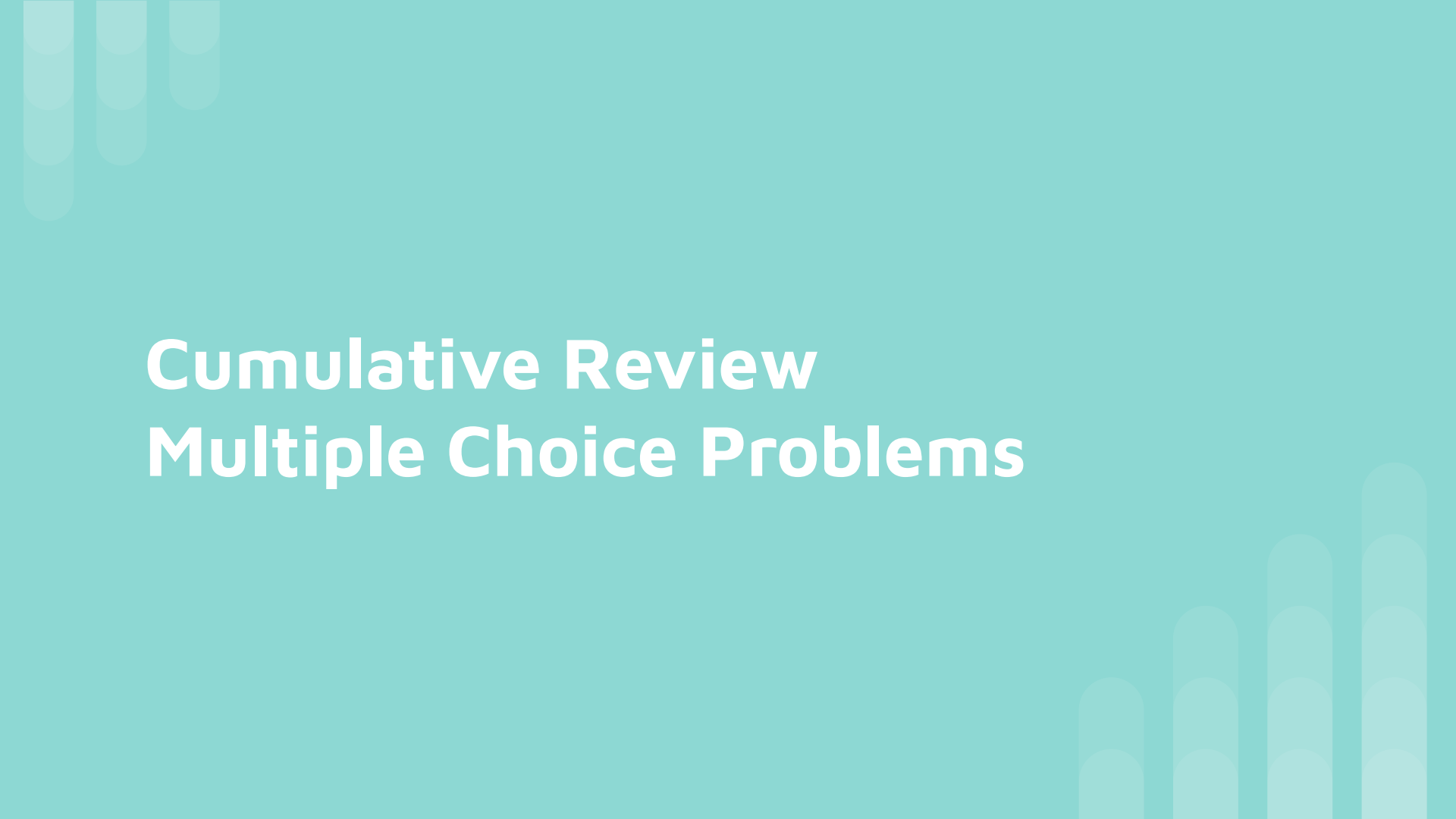
Chapter 20+21 Practice Problems

12. Higher dividend payout policies have a _____ impact on the value of the call and a _____ impact on the value of the put compared to lower dividend payout policies.
- A. negative, negative
 - B. positive, positive
 - C. positive, negative
 - D. negative, positive
 - E. zero, zero
13. The intrinsic value of an at-the-money call option is equal to
- A. the call premium.
 - B. zero.
 - C. the stock price plus the exercise price.
 - D. the striking price.
 - E. none of the above.



Chapter 20+21 Practice Problems

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 - D. the striking price.
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Cumulative Review

Multiple Choice Problems



Cumulative Review

30. Consider two perfectly negatively correlated risky securities A and B. A has an expected rate of return of 10% and a standard deviation of 16%. B has an expected rate of return of 8% and a standard deviation of 12%. The weights of A and B in the global minimum variance portfolio are _____ and _____, respectively.

- a. 0.24; 0.76
- b. 0.50; 0.50
- c. 0.57; 0.43
- d. 0.43; 0.57
- e. 0.76; 0.24



Cumulative Review - Chapter 7

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- a. 0.24; 0.76
- b. 0.50; 0.50
- c. 0.57; 0.43
- d. 0.43; 0.57**
- e. 0.76; 0.24

$$w_A = 12/(16 + 12) = 0.4286; w_B = 1 - 0.4286 = 0.5714.$$



Cumulative Review

1. The bid price of a T-bill in the secondary market is
 - A. the price at which the dealer in T-bills is willing to sell the bill.
 - B. the price at which the dealer in T-bills is willing to buy the bill.
 - C. greater than the asked price of the T-bill.
 - D. the price at which the investor can buy the T-bill.
 - E. never quoted in the financial press.



Cumulative Review - Chapter 2

1. The bid price of a T-bill in the secondary market is
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Cumulative Review

Given the following two stocks A and B

Security	Expected rate of return	Beta
A	0.12	1.2
B	0.14	1.8

If the expected market rate of return is 0.09 and the risk-free rate is 0.05, which security would be considered the better buy and why?

- A. A because it offers an expected excess return of 1.2%.
- B. B because it offers an expected excess return of 1.8%.
- C. A because it offers an expected excess return of 2.2%.
- D. B because it offers an expected return of 14%.
- E. B because it has a higher beta.



Cumulative Review - Chapter 9

Given the following two stocks A and B

Security	Expected rate of return	Beta
A	0.12	1.2
B	0.14	1.8

If the expected market rate of return is 0.09 and the risk-free rate is 0.05, which security would be considered the better buy and why?

- A. A because it offers an expected excess return of 1.2%.
- B. B because it offers an expected excess return of 1.8%.
- C. A because it offers an expected excess return of 2.2%.**
- D. B because it offers an expected return of 14%.
- E. B because it has a higher beta.

CAPM return for A = $0.05 + 1.2(0.09 - 0.05) = \mathbf{0.098}$

CAPM return for B = $0.05 + 1.8(0.09 - 0.05) = \mathbf{0.122}$

A alpha = **0.022**

B alpha = **0.018**

A alpha > B alpha

A is more underpriced than B



Cumulative Review

2. Consider the following three stocks:

<u>Stock</u>	<u>Price</u>	<u>Number of shares outstanding</u>
Stock A	\$40	200
Stock B	\$70	500
Stock C	\$10	600

The price-weighted index constructed with the three stocks is

- A. 30
- B. 40
- C. 50
- D. 60
- E. 70



Cumulative Review - Chapter 2

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Stock C	\$10	600

The price-weighted index constructed with the three stocks is

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- B. 40**
- C. 50
- D. 60
- E. 70

$$(40 + 70 + 10) / 3 = \$40$$



Cumulative Review

3. Consider the following three stocks:

<u>Stock</u>	<u>Price</u>	<u>Number of shares outstanding</u>
Stock A	\$40	200
Stock B	\$70	500
Stock C	\$10	600

The value-weighted index constructed with the three stocks using a divisor of 100 is

- A. 1.2.
- B. 1200.
- C. 490.
- D. 4900.
- E. 49.



Cumulative Review - Chapter 2

3. Consider the following three stocks:

<u>Stock</u>	<u>Price</u>	<u>Number of shares outstanding</u>
Stock A	\$40	200
Stock B	\$70	500
Stock C	\$10	600

The value-weighted index constructed with the three stocks using a divisor of 100 is

- A. 1.2.
- B. 1200.
- C. 490.**
- D. 4900.
- E. 49.

Mkt Cap A = $\$40 * 200 = \$8,000$

Mkt Cap B = $\$70 * 500 = \$35,000$

Mkt Cap C = $\$10 * 600 = \$6,000$

Total Value = $\$49,000 / 100 = \490 ... S+P 500 is a value-weighted index



Cumulative Review

A finding that _____ would provide evidence against the semistrong form of the efficient market theory.

- A. low P/E stocks tend to have positive abnormal returns
- B. trend analysis is worthless in determining stock prices
- C. one can consistently outperform the market by adopting the contrarian approach exemplified by the reversals phenomenon
- D. A and B
- E. A and C



Cumulative Review - Chapter 11

A finding that _____ would provide evidence against the semistrong form of the efficient market theory.

A. low P/E stocks tend to have positive abnormal returns

B. trend analysis is worthless in determining stock prices

C. one can consistently outperform the market by adopting the contrarian approach exemplified by the reversals phenomenon

D. A and B

E. A and C



Cumulative Review

4. The price quotations of Treasury bonds in the Wall Street Journal show an ask price of 104:08 and a bid price of 104:04. As a buyer of the bond, what is the dollar price you expect to pay?

- A. \$1,048.00
- B. \$1,042.50
- C. \$1,044.00
- D. \$1,041.25
- E. \$1,040.40



Cumulative Review - Chapter 2

4. The price quotations of Treasury bonds in the Wall Street Journal show an ask price of 104:08 and a bid price of 104:04. As a buyer of the bond, what is the dollar price you expect to pay?

- A. \$1,048.00
- B. \$1,042.50**
- C. \$1,044.00
- D. \$1,041.25
- E. \$1,040.40

104 and 8/32 % of total face value... $1,000 \times 1.0425 = \$1,042.50$

The colon notation indicates the numerator of $x / 32$. Practically, having prices vary by $1/32$ instead of $1/100$ makes trade computing easier/quicker.



Cumulative Review

5. An investor purchases one municipal and one corporate bond that pay rates of return of 8% and 10%, respectively. If the investor is in the 20% marginal tax bracket, his or her after-tax rates of return on the municipal and corporate bonds would be _____ and _____, respectively.

- A. 8% and 10%
- B. 8% and 8%
- C. 6.4% and 8%
- D. 6.4% and 10%
- E. 10% and 10%



Cumulative Review - Chapter 2

5. An investor purchases one municipal and one corporate bond that pay rates of return of 8% and 10%, respectively. If the investor is in the 20% marginal tax bracket, his or her after-tax rates of return on the municipal and corporate bonds would be _____ and _____, respectively.

- A. 8% and 10%
- B. 8% and 8%**
- C. 6.4% and 8%
- D. 6.4% and 10%
- E. 10% and 10%

$$\text{Taxed Yield} = r_C(1-t) = 0.10(1-0.2) = 0.08$$

Remember! When trying to find muni parity ... $r_M = r_C(1-t)$



Cumulative Review

Consider two bonds, F and G. Both bonds presently are selling at their par value of \$1,000. Each pays interest of \$90 annually. Bond F will mature in 15 years while bond G will mature in 26 years. If the yields to maturity on the two bonds change from 9% to 10%,

_____.

- A. both bonds will increase in value, but bond F will increase more than bond G
- B. both bonds will increase in value, but bond G will increase more than bond F
- C. both bonds will decrease in value, but bond F will decrease more than bond G
- D. both bonds will decrease in value, but bond G will decrease more than bond F
- E. none of the above



Cumulative Review - Chapter 14

Bond F will mature in 15 years while bond G will mature in 26 years. If the yields to maturity on the two bonds change from 9% to 10%

A. both bonds will increase in value, but bond F will increase more than bond G

B. both bonds will increase in value, but bond G will increase more than bond F

C. both bonds will decrease in value, but bond F will decrease more than bond G

D. both bonds will decrease in value, but bond G will decrease more than bond F

E. none of the above

$$\sum_{t=1}^T \frac{\text{Coupon}}{(1+r)^t} + \frac{\text{Par value}}{(1+r)^T}$$

Increase in r increases denominator, so both bonds will decrease in value

$1+r$ is to the power of t so the bond with bigger T will decrease more



Cumulative Review

10. You sold JCP stock short at \$80 per share. Your losses could be minimized by placing a

- A. limit-sell order.
- B. limit-buy order.
- C. stop-buy order.
- D. day-order.
- E. None of the options



Cumulative Review - Chapter 3

10. You sold JCP stock short at \$80 per share. Your losses could be minimized by placing a

A. limit-sell order.

B. limit-buy order.

C. stop-buy order.

D. day-order.

E. None of the options



Cumulative Review

A 7.5% coupon bond with an ask price of 100:00 pays interest every 182 days. If the bond paid interest 62 days ago, the invoice price of the bond would be

- A. 1,011.67
- B. 1,012.35
- C. 1,012.77
- D. 1,011.98
- E. 1,012.15



Cumulative Review - Chapter 14

A 7.5% coupon bond with an ask price of 100:00 pays interest every 182 days. If the bond paid interest 62 days ago, the invoice price of the bond would be

A. 1,011.67

B. 1,012.35

C. 1,012.77

D. 1,011.98

E. 1,012.15

$$\begin{aligned}\text{Invoice price} &= \text{flat price} + \text{accrued interest} \\ &= 1000 + (62/182) * 0.075 * \mathbf{0.5} * 1000 \\ &= 1012.77\end{aligned}$$

Remember to multiply accrued interest by 0.5 because coupon payments are semiannual!



Cumulative Review

14. Assume you purchased 200 shares of GE common stock on margin at \$70 per share from your broker. If the initial margin is 55%, how much did you borrow from the broker?
- A. \$6,000
 - B. \$4,000
 - C. \$7,700
 - D. \$7,000
 - E. \$6,300



Cumulative Review - Chapter 3

14. Assume you purchased 200 shares of GE common stock on margin at \$70 per share from your broker. If the initial margin is 55%, how much did you borrow from the broker?

A. \$6,000

B. \$4,000

C. \$7,700

D. \$7,000

E. \$6,300

$$200 \times \$70 \times (1 - 0.55) = \$6300$$



Cumulative Review

Consider an 8% coupon bond selling for \$1000 with three years until maturity making annual coupon payments. The interest rates in the next three years will be, with certainty, $r_1 = 8\%$, $r_2 = 10\%$, and $r_3 = 12\%$. What is the realized compound yield?

- a. 10.4%
- b. 9.9%
- c. 12.3%
- d. 8.2%
- e. 13.5%



Cumulative Review - Chapter 14

Consider an 8% coupon bond selling for \$1000 with three years until maturity making annual coupon payments. The interest rates in the next three years will be, with certainty, $r_1 = 8\%$, $r_2 = 10\%$, and $r_3 = 12\%$. What is the realized compound yield?

- a. 10.4%
- b. 9.9%
- c. 12.3%
- d. 8.2%**
- e. 13.5%

$$\begin{aligned}\text{Total payout} &= 80 \times 1.1 \times 1.12 + 80 \times 1.12 + 80 + 1000 = 1268.16 \\ \text{initial investment} \times (1+r)^t &= \text{total payout} \\ 1000 \times (1+r)^3 &= 1268.16 \\ (1+r)^3 &= 1.268 \\ 1+r &= 1.082 \\ r &= 0.082\end{aligned}$$



Cumulative Review

18. You purchased 300 shares of common stock on margin for \$60 per share. The initial margin is 60% and the stock pays no dividend. What would your rate of return be if you sell the stock at \$45 per share? Ignore interest on margin.
- A. 25.00%
 - B. -33.33%
 - C. 44.31%
 - D. -41.67%
 - E. -54.22%



Cumulative Review - Chapter 3

18. You purchased 300 shares of common stock on margin for \$60 per share. The initial margin is 60% and the stock pays no dividend. What would your rate of return be if you sell the stock at \$45 per share? Ignore interest on margin.

- A. 25.00%
- B. -33.33%
- C. 44.31%
- D. -41.67%
- E. -54.22%

Initial investment = $300(60)(.6) = \$10,800$
 $300(\$60) = 18000 \times (0.40) = \7200 loan
After selling stock and repaying the loan:
 $13500 - 7200 = 6300$
Return = $(6300 - 10800) / 10800 = -41.67\%$



Cumulative Review

6. Diversified Portfolios had year-end assets of \$279,000,000 and liabilities of \$43,000,000. If Diversified's NAV was \$42.13, how many shares must have been held in the fund?
- A. 43,000,000
 - B. 6,488,372
 - C. 5,601,709
 - D. 1,182,203



Cumulative Review - Chapter 4

6. Diversified Portfolios had year-end assets of \$279,000,000 and liabilities of \$43,000,000. If Diversified's NAV was \$42.13, how many shares must have been held in the fund?

A. 43,000,000

B. 6,488,372

C. 5,601,709

D. 1,182,203

$$(279,000,000 - 43,000,000) / 42.13 = 5601708.996$$



Cumulative Review

11. Closed end funds are frequently issued at a _____ to NAV and subsequently trade at a _____ to NAV.

- a. Discount, discount
- b. Discount, premium
- c. Premium, premium
- d. Premium, discount
- e. No consistent relationship has been observed



Cumulative Review - Chapter 4

11. Closed end funds are frequently issued at a _____ to NAV and subsequently trade at a _____ to NAV.

- a. Discount, discount
- b. Discount, premium
- c. Premium, premium
- d. Premium, discount**
- e. No consistent relationship has been observed



Cumulative Review

23. Investors' Choice Fund had NAV per share of \$37.25 on January 1, 2012. On December 31 of the same year the fund's rate of return for the year was 17.3%. Income distributions were \$1.14, and the fund had capital gain distributions of \$1.35. Without considering taxes and transactions costs, what ending NAV would you calculate for Investors' Choice?

- a. \$41.20
- b. \$33.88
- c. \$43.69
- d. \$42.03
- e. \$46.62



Cumulative Review - Chapter 4

23. Investors' Choice Fund had NAV per share of \$37.25 on January 1, 2012. On December 31 of the same year the fund's rate of return for the year was 17.3%. Income distributions were \$1.14, and the fund had capital gain distributions of \$1.35. Without considering taxes and transactions costs, what ending NAV would you calculate for Investors' Choice?

- a. **\$41.20**
- b. \$33.88
- c. \$43.69
- d. \$42.03
- e. \$46.62

$$.173 = (P - \$37.25 + 1.14 + 1.35)/\$37.25 \rightarrow P = \$41.20$$



Cumulative Review

1. You purchased a share of stock for \$68. One year later you received \$3.00 as a dividend and sold the share for \$74.50. What was your holding-period return?

- A. 12.5%
- B. 14.0%
- C. 13.6%
- D. 11.8%
- E. 4.20%



Cumulative Review - Chapter 5

1. You purchased a share of stock for \$68. One year later you received \$3.00 as a dividend and sold the share for \$74.50. What was your holding-period return?

- A. 12.5%
- B. 14.0%**
- C. 13.6%
- D. 11.8%
- E. 4.20%

$$\text{HPR} = (74.50 - 68.00 + 3.00) / 68.00 = 0.1397 \text{ or } 14\%$$



Cumulative Review

2. You have been given this probability distribution for the holding-period return for KMP stock:

<u>State of the Economy</u>	<u>Probability</u>	<u>HPR</u>
Boom	.30	18%
Normal growth	.50	12%
Recession	.20	- 5%

What is the expected holding-period return for KMP stock?

- A. 10.40%
- B. 9.32%
- C. 11.63%
- D. 11.54%
- E. 10.88%



Cumulative Review - Chapter 5

2. You have been given this probability distribution for the holding-period return for KMP stock:

<u>State of the Economy</u>	<u>Probability</u>	<u>HPR</u>
Boom	.30	18%
Normal growth	.50	12%
Recession	.20	- 5%

What is the expected holding-period return for KMP stock?

- A. 10.40%
- B. 9.32%
- C. 11.63%
- D. 11.54%
- E. 10.88%

$$\text{HPR} = .30 (18\%) + .50 (12\%) + .20 (-5\%) = 10.4\%$$



Cumulative Review

3. You have been given this probability distribution for the holding-period return for KMP stock:

<u>State of the Economy</u>	<u>Probability</u>	<u>HPR</u>
Boom	.30	18%
Normal growth	.50	12%
Recession	.20	- 5%

What is the expected variance for KMP stock?

- A. 66.04%
- B. 69.96%
- C. 77.04%
- D. 63.72%
- E. 78.45%



Cumulative Review

A zero-coupon bond has a yield to maturity of 11% and a par value of \$1,000. If the bond matures in 27 years, the bond should sell for a price of _____ today.

- A. \$59.74
- B. \$501.87
- C. \$513.16
- D. \$483.49
- E. none of the above



Cumulative Review - Chapter 14

A zero-coupon bond has a yield to maturity of 11% and a par value of \$1,000. If the bond matures in 27 years, the bond should sell for a price of _____ today.

A. \$59.74

$$\text{Bond value} = \text{Coupon payment} \left(\frac{1}{r} \left(1 - \frac{1}{(1+r)^T} \right) \right) + \text{Par value} \frac{1}{(1+r)^T}$$

B. \$501.87

$$= 0 + 1000(1/(1+0.11)^{27}) = 59.74$$

C. \$513.16

D. \$483.49

E. none of the above



Cumulative Review

11. A portfolio has an expected rate of return of 0.15 and a standard deviation of 0.15. The risk-free rate is 6%. An investor has the following utility function: $U = E(r) - (A/2)s^2$. Which value of A makes this investor indifferent between the risky portfolio and the risk-free asset?

- a. 5
- b. 6
- c. 7
- d. 8



Cumulative Review - Chapter 6

11. A portfolio has an expected rate of return of 0.15 and a standard deviation of 0.15. The risk-free rate is 6%. An investor has the following utility function: $U = E(r) - (A/2)s^2$. Which value of A makes this investor indifferent between the risky portfolio and the risk-free asset?

- a. 5
- b. 6
- c. 7
- d. 8

$$0.06 = 0.15 - A/2(0.15)^2;$$

$$0.06 - 0.15 = -A/2(0.0225);$$

$$-0.09 = -0.01125A; A = 8;$$

$$U = 0.15 - 8/2(0.15)^2 = 6\%; U(R_f) = 6\%.$$



Cumulative Review

26. Which of the following statements regarding the capital allocation line (CAL) is **false**?

- a. The CAL shows risk-return combinations
- b. The slope of the CAL equals the increase in the expected return of the complete portfolio per unit of additional standard deviation
- c. The slope of the CAL is also called the reward-to-volatility ratio
- d. The CAL is also called the efficient frontier of risky assets in the absence of a risk-free asset.



Cumulative Review - Chapter 6

26. Which of the following statements regarding the capital allocation line (CAL) is **false**?

- a. The CAL shows risk-return combinations
- b. The slope of the CAL equals the increase in the expected return of the complete portfolio per unit of additional standard deviation
- c. The slope of the CAL is also called the reward-to-volatility ratio
- d. **The CAL is also called the efficient frontier of risky assets in the absence of a risk-free asset.**

The CAL consists of combinations of a risky asset and a risk-free asset whose slope is the reward-to-volatility ratio; thus, all statements except D are true.



Cumulative Review

30. An investor invests 40% of his wealth in a risky asset with an expected rate of return of 0.17 and a variance of 0.08 and 60% in a T-bill that pays 4.5%. His portfolio's expected return and standard deviation are _____ and _____, respectively.

- a. 0.114; 0.126
- b. 0.087; 0.068
- c. 0.095; 0.113
- d. 0.087; 0.124
- e. None of the options



Cumulative Review - Chapter 6

30. An investor invests 40% of his wealth in a risky asset with an expected rate of return of 0.17 and a variance of 0.08 and 60% in a T-bill that pays 4.5%. His portfolio's expected return and standard deviation are _____ and _____, respectively.

- a. 0.114; 0.126
- b. 0.087; 0.068
- c. **0.095; 0.113**
- d. 0.087; 0.124
- e. None of the options

$$E(r_p) = 0.4(17\%) + 0.6(4.5\%) = 9.5\%; s_p = 0.4(0.08)^{1/2} = 11.31\%.$$



Cumulative Review

34. Which of the following cannot lie on the efficient frontier as described by Markowitz?

<u>Portfolio</u>	<u>Expected Return</u>	<u>Standard Deviation</u>
W	9%	21%
X	5%	7%
Y	15%	36%
Z	12%	15%

- a. Only portfolio W cannot lie on the efficient frontier
- b. Only portfolio X cannot lie on the efficient frontier
- c. Only portfolio Y cannot lie on the efficient frontier
- d. Only portfolio Z cannot lie on the efficient frontier
- e. Cannot tell from the information given.



Cumulative Review - Chapter 7

34. Which of the following cannot lie on the efficient frontier as described by Markowitz?

<u>Portfolio</u>	<u>Expected Return</u>	<u>Standard Deviation</u>
W	9%	21%
X	5%	7%
Y	15%	36%
Z	12%	15%

W has a higher standard deviation than Z with a lower expected return and would lie below the efficient frontier as described by Markowitz

- a. Only portfolio W cannot lie on the efficient frontier
- b. Only portfolio X cannot lie on the efficient frontier
- c. Only portfolio Y cannot lie on the efficient frontier
- d. Only portfolio Z cannot lie on the efficient frontier
- e. Cannot tell from the information given.



Cumulative Review

46. Security X has expected return of 12% and standard deviation of 18%. Security Y has expected return of 15% and standard deviation of 26%. If the two securities have a correlation coefficient of 0.7, what is their covariance?

- a. 0.038
- b. 0.070
- c. 0.018
- d. 0.033
- e. 0.054



Cumulative Review - Chapter 7

46. Security X has expected return of 12% and standard deviation of 18%. Security Y has expected return of 15% and standard deviation of 26%. If the two securities have a correlation coefficient of 0.7, what is their covariance?

- a. 0.038
- b. 0.070
- c. 0.018
- d. **0.033**
- e. 0.054

$$\text{Cov}(r_X, r_Y) = (0.7)(.18)(.26) = .0327$$